SEPARATEDNESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1916-1922, following the statement at 1910-1914.

Editorial note. The parenthetical condition checks (diagonal lft, qcqs fibre product) are stated with 'Some details omitted'; the same facts are anchored by named lemmas in cards L26359/L26362; flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1921 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Closed immersions appear at a finite stage. lemma-descend-closed-immersion (same chapter, card L26349, cited at ~1916-1918). Applied to the diagonal Delta_{X_0/Y_0}: X_0 -> X_0 x_{Y_0} X_0.

Source reference. Original lemma 0850: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1887–1887. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If f is separated, then f_i is separated for some $i \geq 0$.*

Proof. Apply Lemma 0850 to the diagonal morphism $\Delta_{X_0/Y_0} : X_0 \to X_0 \times_{Y_0} X_0$. (Diagonal morphisms are locally of finite type and the fibre product $X_0 \times_{Y_0} X_0$ is quasi-compact and quasi-separated. Some details omitted.) □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.